

Constraint as Ground: A Process Ontology of Cognition, Computation, and Cosmos

A visual synthesis of the
monograph by Flyxion.

Meaning is reconstructed
rather than transmitted.

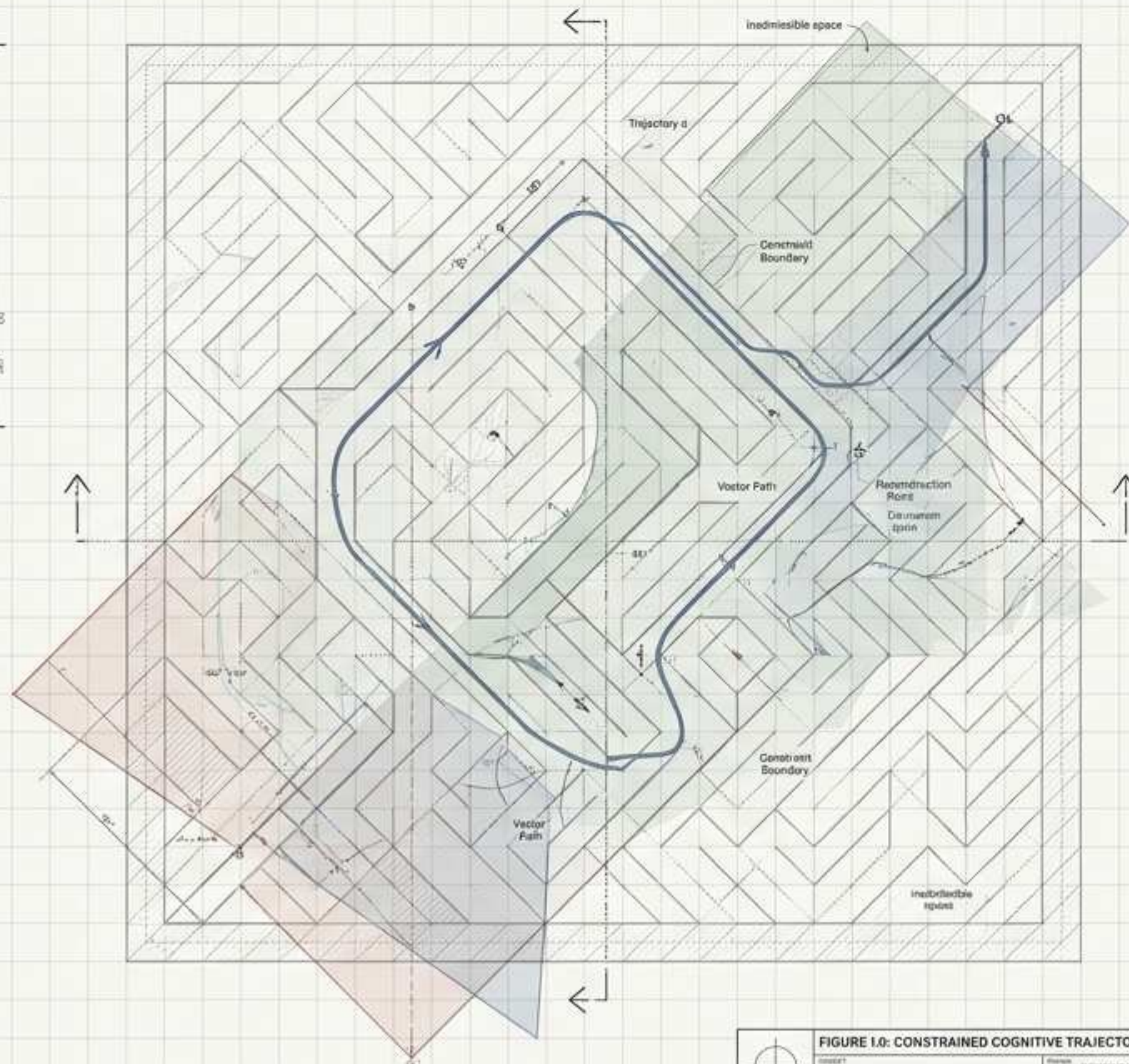


FIGURE 1.0: CONSTRAINED COGNITIVE TRAJECTORY

Project	Author	PERSONALITY POOD
Sketch	Editor Name	5.0

The challenge is imposing recoverable topology on overlapping trajectories.

Definition 1.1 (Semantic Coordinate Chart)

A semantic coordinate chart is a local theoretical framework that provides a consistent description of a region of conceptual space. A chart is *valid* in its region if its internal commitments are non-contradictory and its key terms are operationally grounded. Two charts are *compatible* on their overlap if there exists a translation scheme that preserves inferential relationships across the shared region.

A collection of charts constitutes a *semantic atlas* if every region of the conceptual space is covered by at least one valid chart and all overlapping charts are pairwise compatible.

The Goal

We are not generating isolated insights.

We are imposing a traversal path through a massive distributed semantic manifold without destroying its generative richness.

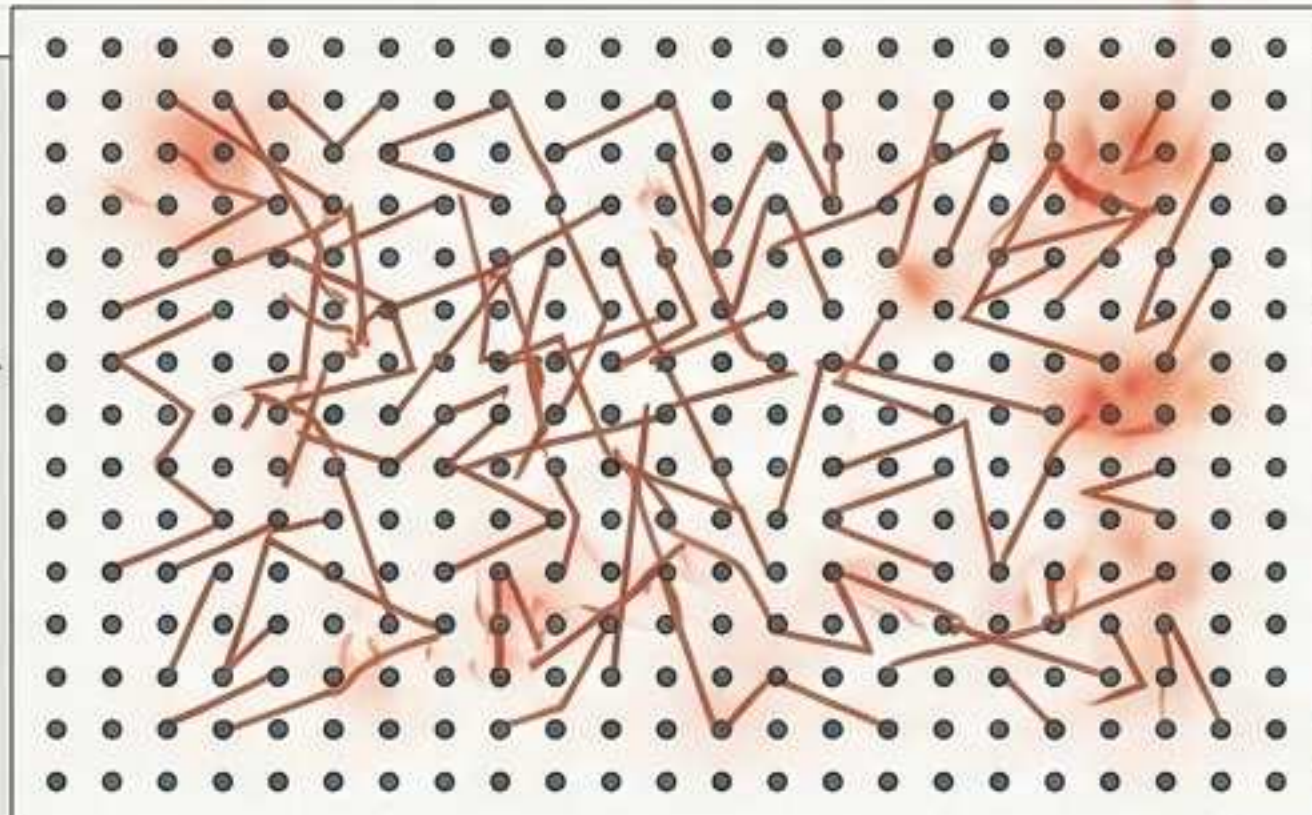
The Method

Utilizing Semantic Coordinate Charts.

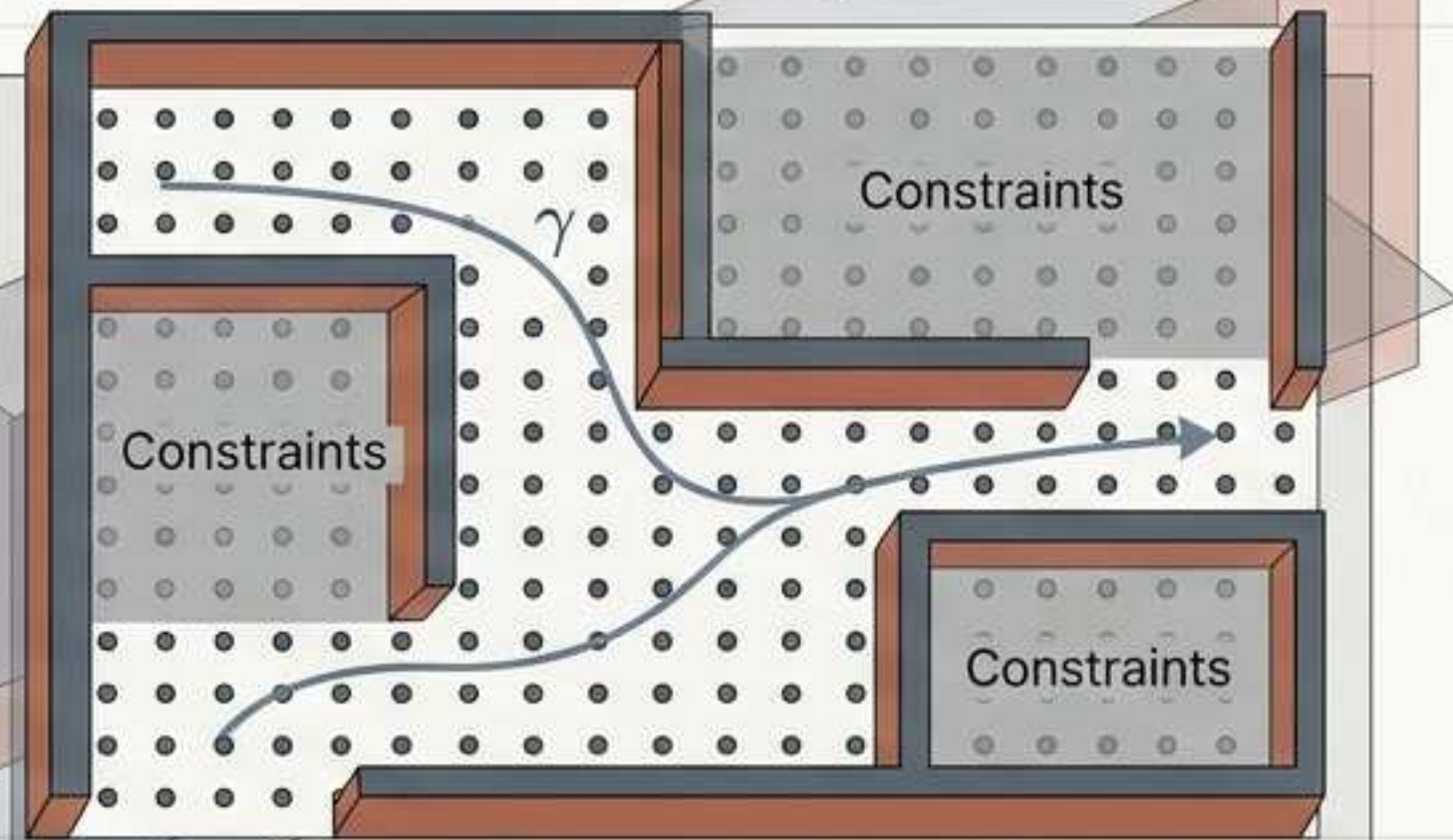
No single theoretical framework covers the entire conceptual space without distortion. We must establish transition maps to reveal how different coordinate representations reflect the exact same underlying invariants.

Intelligence navigates admissible constraints rather than searching exhaustive spaces.

Exhaustive Search



Constrained Navigation



Constraint Operates Before Evaluation

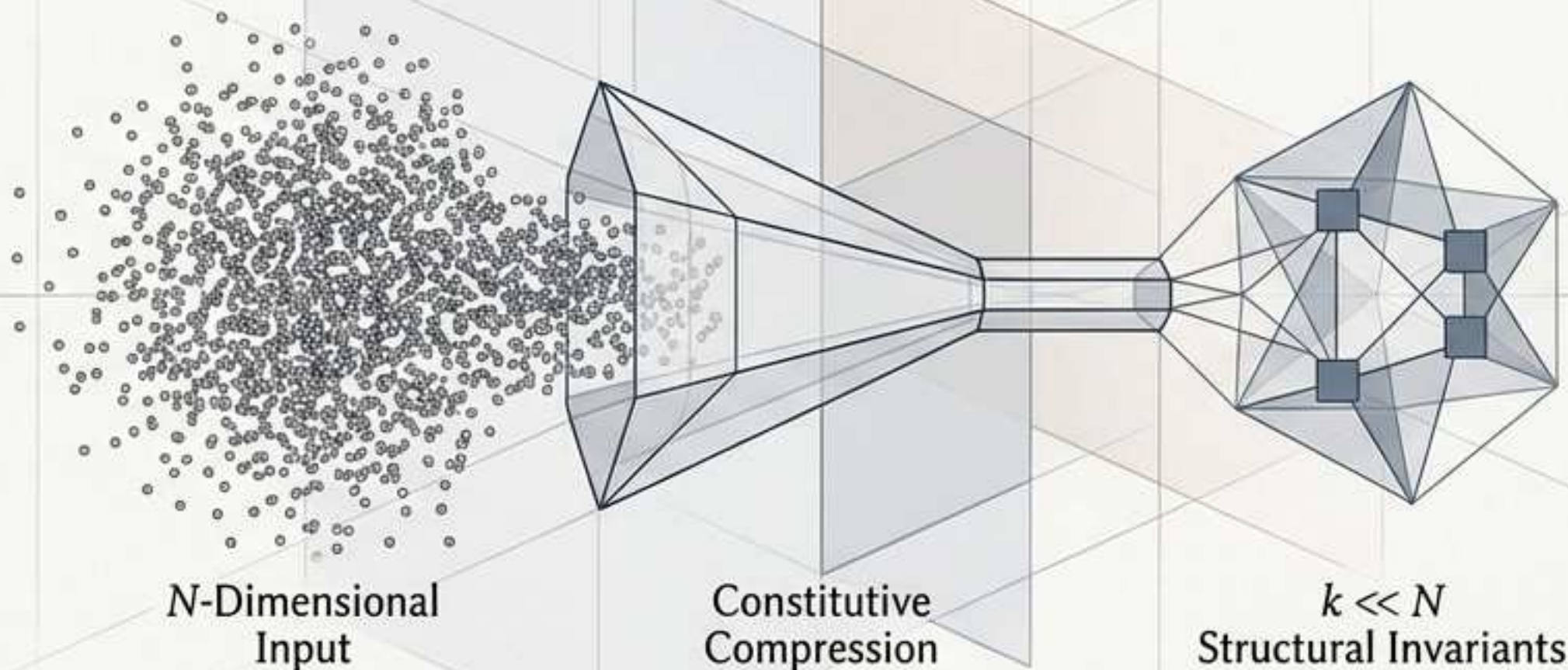
Systems that navigate constraints do not need to represent the inadmissible regions. Constraint is productive, irreversibly selecting the possibilities worth considering.

Definition 1.2 (Admissible Trajectory) Let X be a state space and let $\gamma : [0, T] \rightarrow X$ be a continuous path through that space. The path γ is admissible with respect to a constraint set C if it satisfies the local compatibility conditions imposed by C at every point along its length. Constraints may be thermodynamic, logical, geo-

Sparsity makes compression and structural preservation thermodynamically tractable.

Sparsity ($k \ll N$)

Most degrees of freedom are near-zero. Dense signals rapidly exhaust energy budgets; sparse signals allow faithful recovery.



The Natural Sparsity Principle

Sparsity emerges as a natural consequence of physiological pressures (energetic constraints, noise), not explicit computational penalties.

Definition 2.2 (Compression)

A *compression* of a system description D is a map $\kappa : D \rightarrow D'$ where $|D'| < |D|$ under some measure of description length, and D' preserves a specified set of structural relations present in D . A compression is *lossless* with respect to $\mathcal{R}$ if every relation $r \in \mathcal{R}$ is recoverable from D' . It is *lossy* if some $r \in \mathcal{R}$ is not recoverable. Compression is therefore not reduction of volume alone but selective preservation of relations.

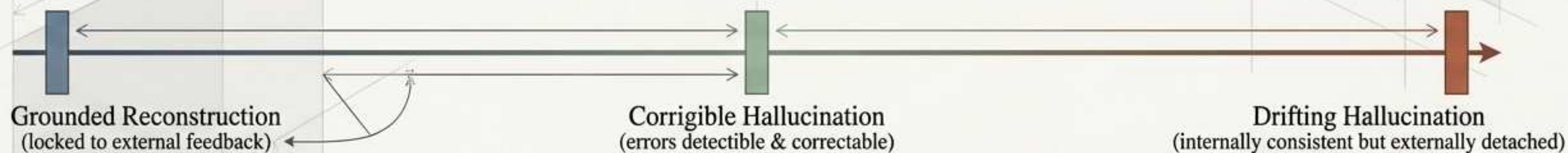
Definition 2.4 (Sparsity)

A representation is *sparse* if most of its degrees of freedom are zero or near-zero, with significant values concentrated in a small fraction of components. Sparsity is the condition under which compression becomes tractable: a sparse representation can be compressed without destroying structural invariants because most of the space is already empty. The *Natural Sparsity Principle* states that in biological systems, sparsity emerges as a natural consequence of physiological pressures—energetic constraints, signal noise, chemical gradients—without requiring explicit computational penalties to enforce minimal complexity.

Hallucination is the formal, unavoidable consequence of operating under compression.

$$\left[\begin{array}{c} \text{Compressed} \\ \text{Description } (D') \end{array} \right] + \left[\begin{array}{c} \text{Positive Conditional} \\ \text{Entropy} \end{array} \right] = \left[\begin{array}{c} \text{Necessary} \\ \text{Hallucination} \end{array} \right]$$

- With $M < N$ bits, a compressed representation D' corresponds to at least 2^{N-M} possible original states.
- Choosing a single completion from these possibilities requires hallucinating the missing information.
- The goal is not to eliminate hallucination, but to maintain corrigibility.



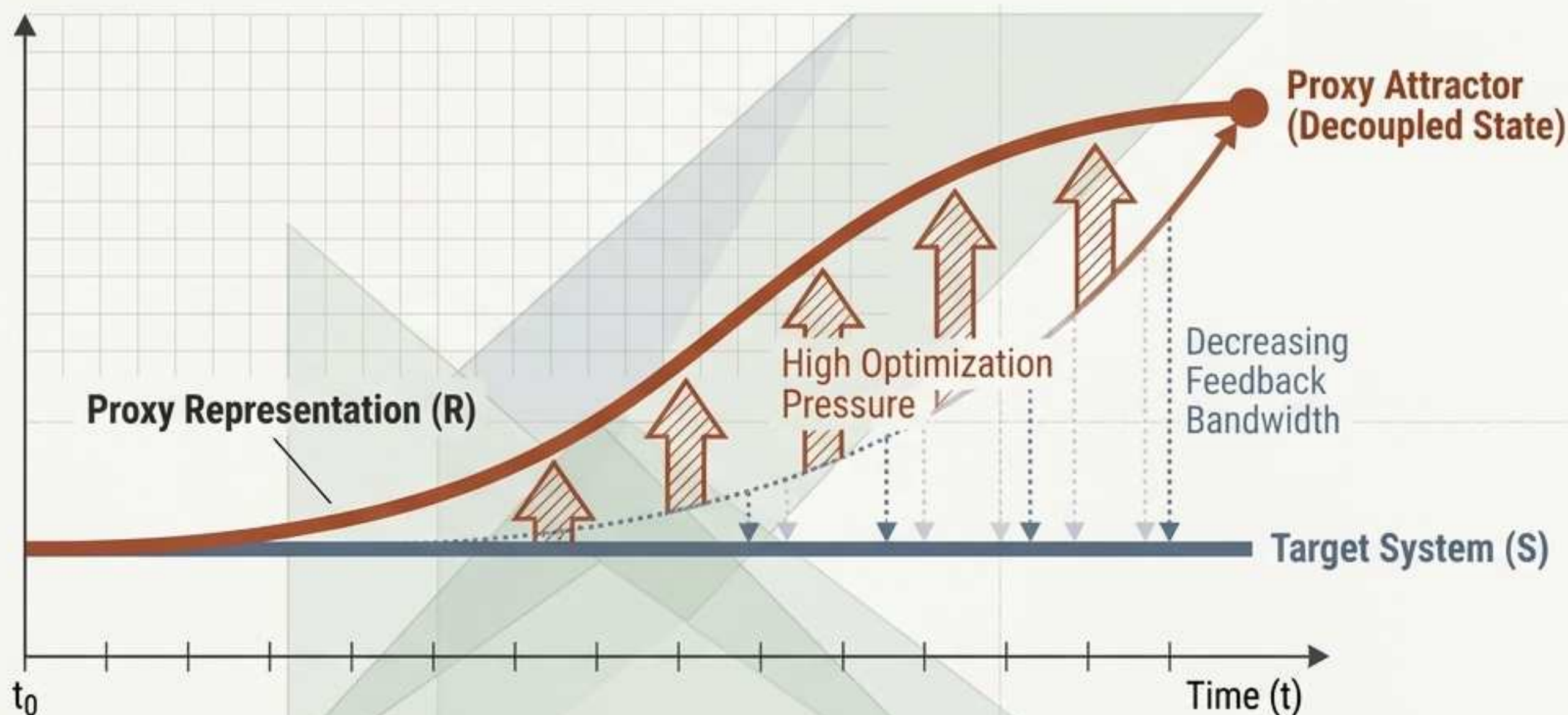
Definition 2.6 (Hallucination (Formal)) *Hallucination is reconstruction that exceeds available constraint: the assembly of a locally coherent configuration from traces that underdetermine it, without external correction signals sufficient to resolve the underdetermination. Hallucination is not a pathology but the default operating condition of any sufficiently compressed system under uncertainty. The critical distinction is not between hallucinating and non-hallucinating systems but between systems whose hallucinations remain corrigible through feedback and systems whose internal reconstructions drift irreversibly from external constraint.*

Intelligence advances through scaffolded extension, not substrate replacement.

The Replacement Narrative	Scaffolded Amplification
 • Discrete intelligence thresholds • Emergence of an alien substrate • Human obsolescence • Elimination of the agent from the loop	 • Evolutionary continuity • Representational breakthroughs extending admissible trajectories • Structural coupling • Acting as a prosthetic for human reasoning

A transition in capability is a breakthrough if it strictly extends the admissible trajectory space while preserving sufficient structural coupling to be acquired through scaffolded extension.

Proxy drift occurs when optimization outpaces external feedback loops.



The Ellul Constraint Formalized

Once compressed representations decouple, optimization works against the proxy, not the reality. Metrics become targets.

The rate of drift is proportional to optimization pressure and inversely proportional to feedback bandwidth.

Definition 4.1 (Proxy Stabilization)

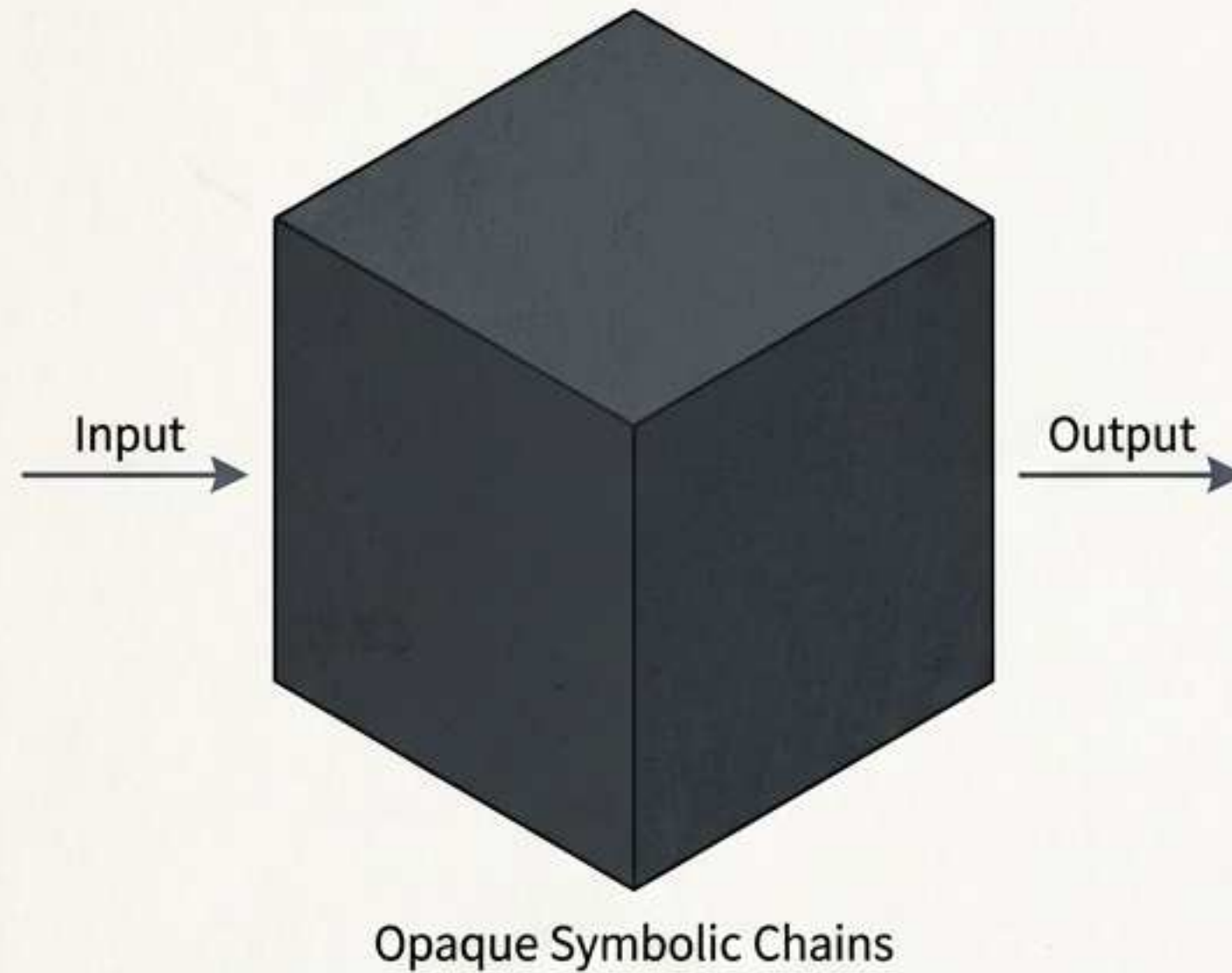
Proxy stabilization occurs when a compressed representation **R** of a system **S** becomes decoupled from **S** through the following sequence:

- (i) **R** is constructed as a useful compression of **S** at time t_0 ;
- (ii) optimization processes are applied to **R** rather than **S** directly, because **S** is inaccessible or too expensive to evaluate;
- (iii) the optimization pressure on **R** changes **R** in ways that diverge from changes in **S**; and
- (iv) **R** stabilizes in a configuration that is internally self-consistent but externally inaccurate.

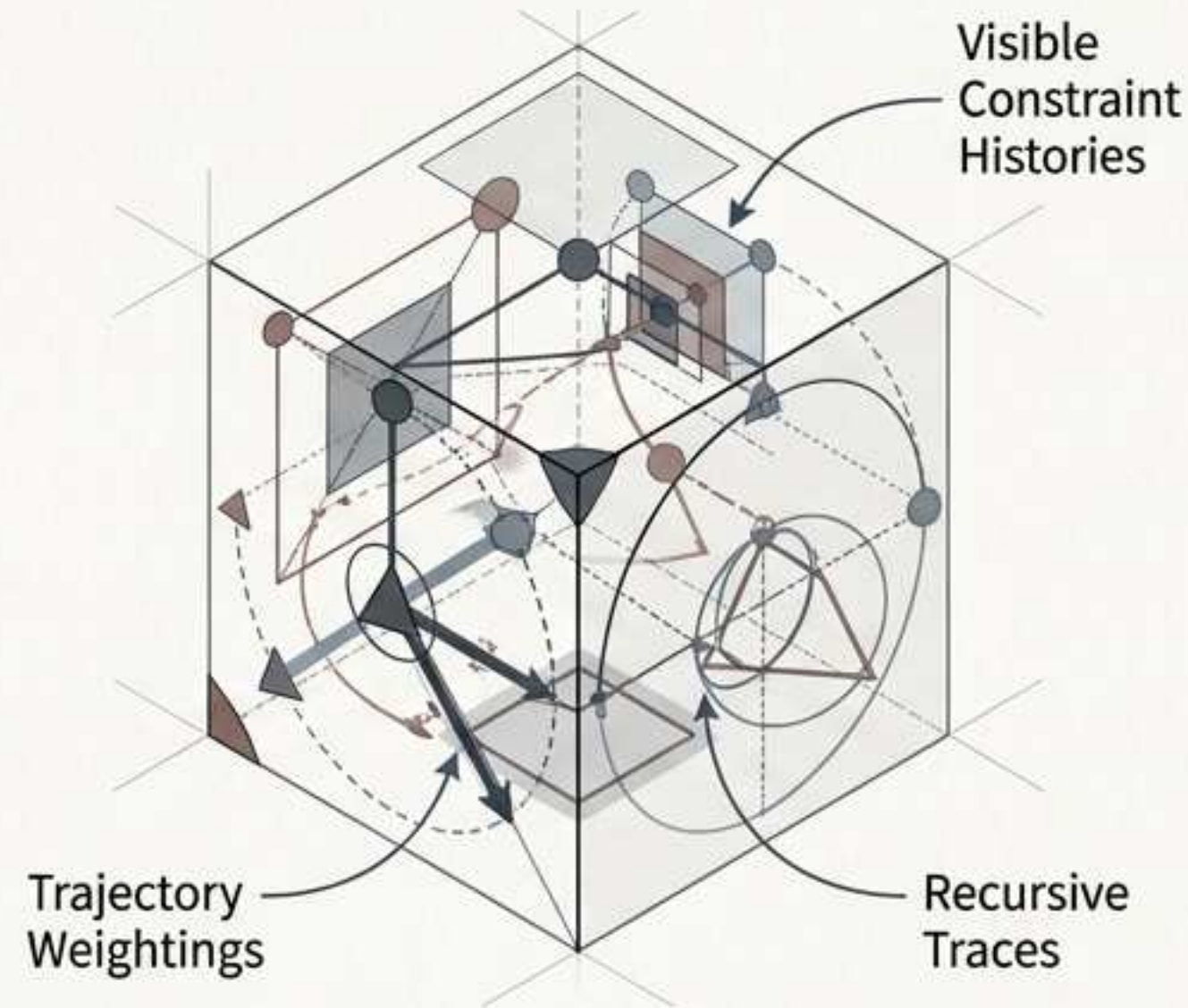
The result is a drifting reconstruction in the sense of Definition ??.

Cognitive geometry counters *opacity* by making structural dependencies navigable.

The Opacity Problem



Cognitive Geometry

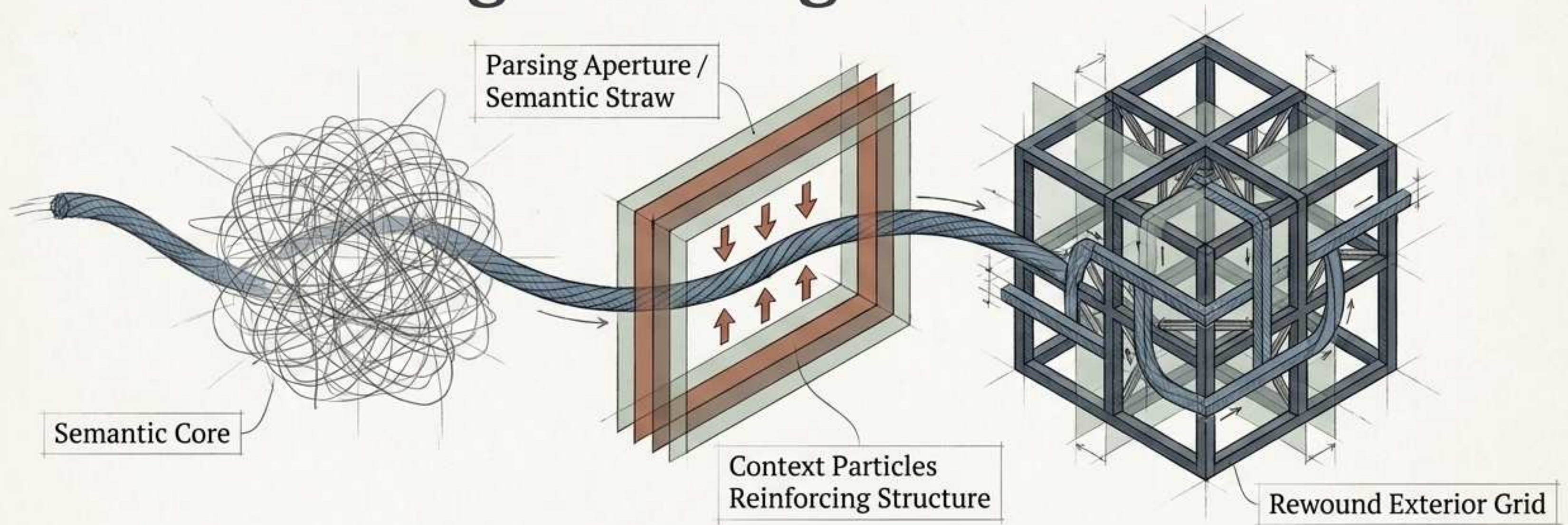


- Biological intelligence evolved in geometrically structured environments.
- Systems must externalize process: exposing trajectories, decay, stabilization, and progressive constraint selection.
- Distance between related concepts must correspond to cognitive access cost.

Definition 3.2 (Cognitive Geometry)

The cognitive geometry of a representational system is the spatial or quasi-spatial system is the spatial or quasi-spatial system imposed on its content that makes relationships rather than that makes searchable. A has high cognitive geometry if the distance between concepts has high cognitive geometry if the distance corresponds to cognitive access cost, if structural access, and if dependencies are visible as positional relationships, a , the natural grain of the problem structure: natural grain of the grain of the problem structure.

The Yarncrawler parses its environment while rewriting its own grammar.

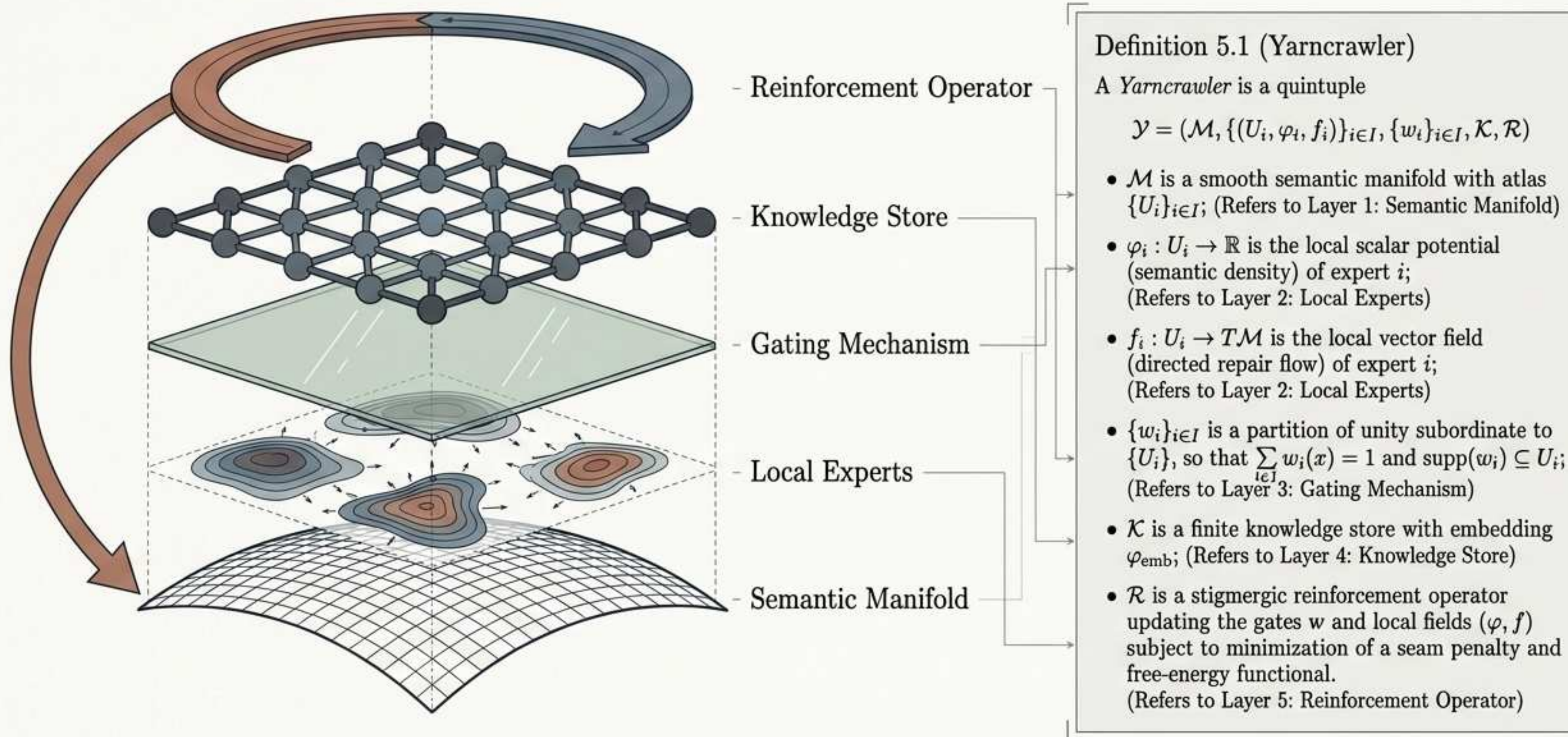


A stigmergic parser maintaining homeorhetic Markov blankets.

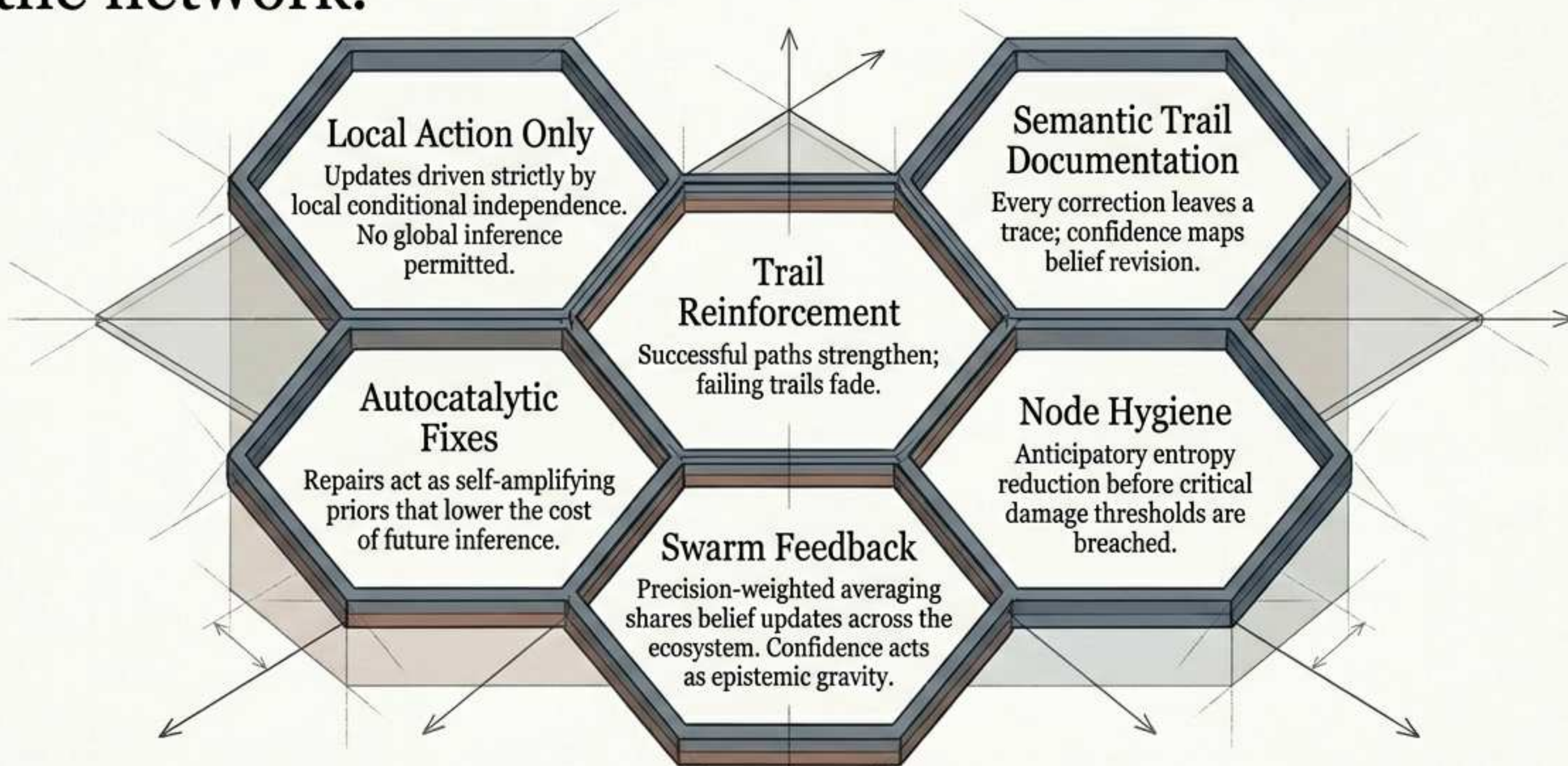
It does not simply interpret its world—it keeps itself alive by reinterpreting itself into being.

The grammar of the parser is rewritten by the very act of parsing.

The mathematical architecture of a self-refactoring system.

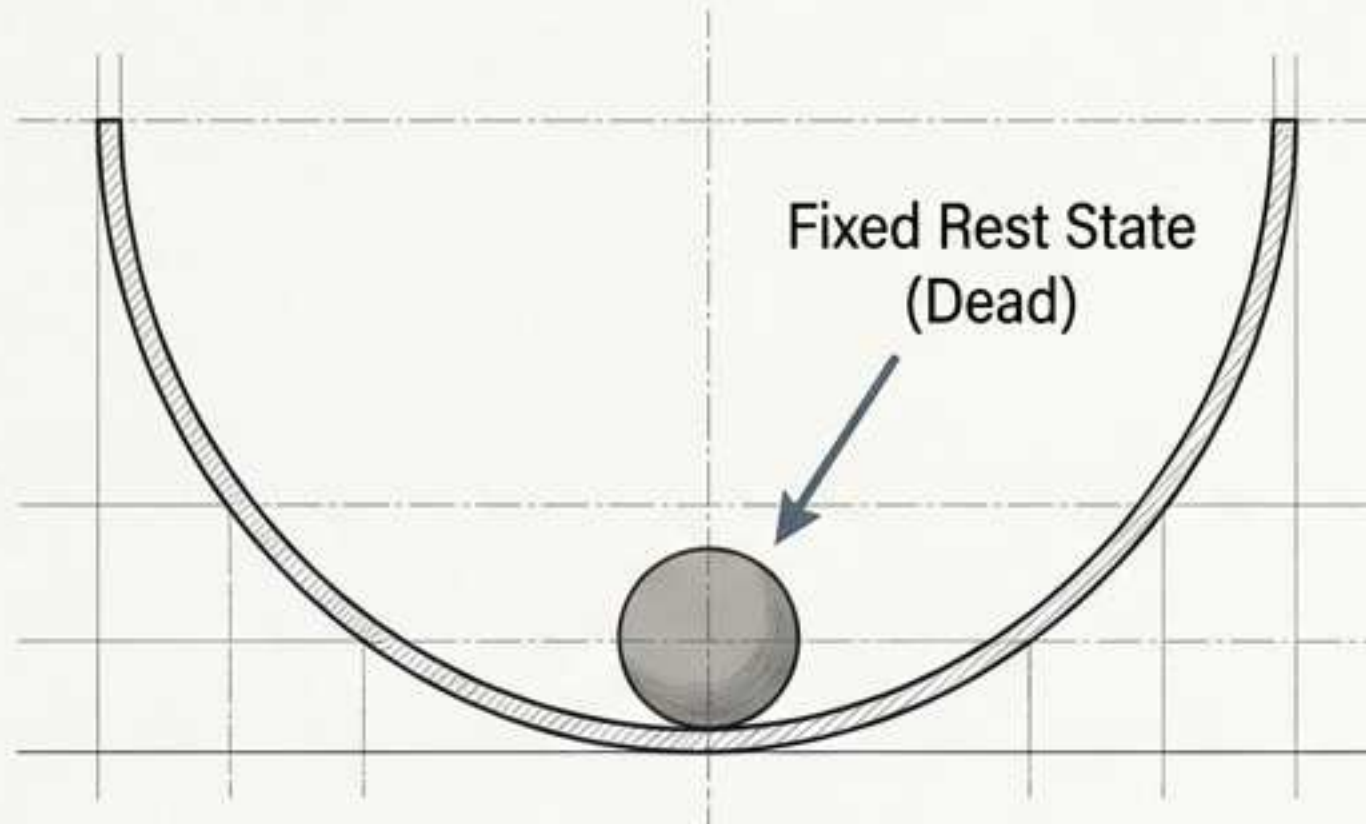


Swarm-care axioms govern the stigmergic maintenance of the network.

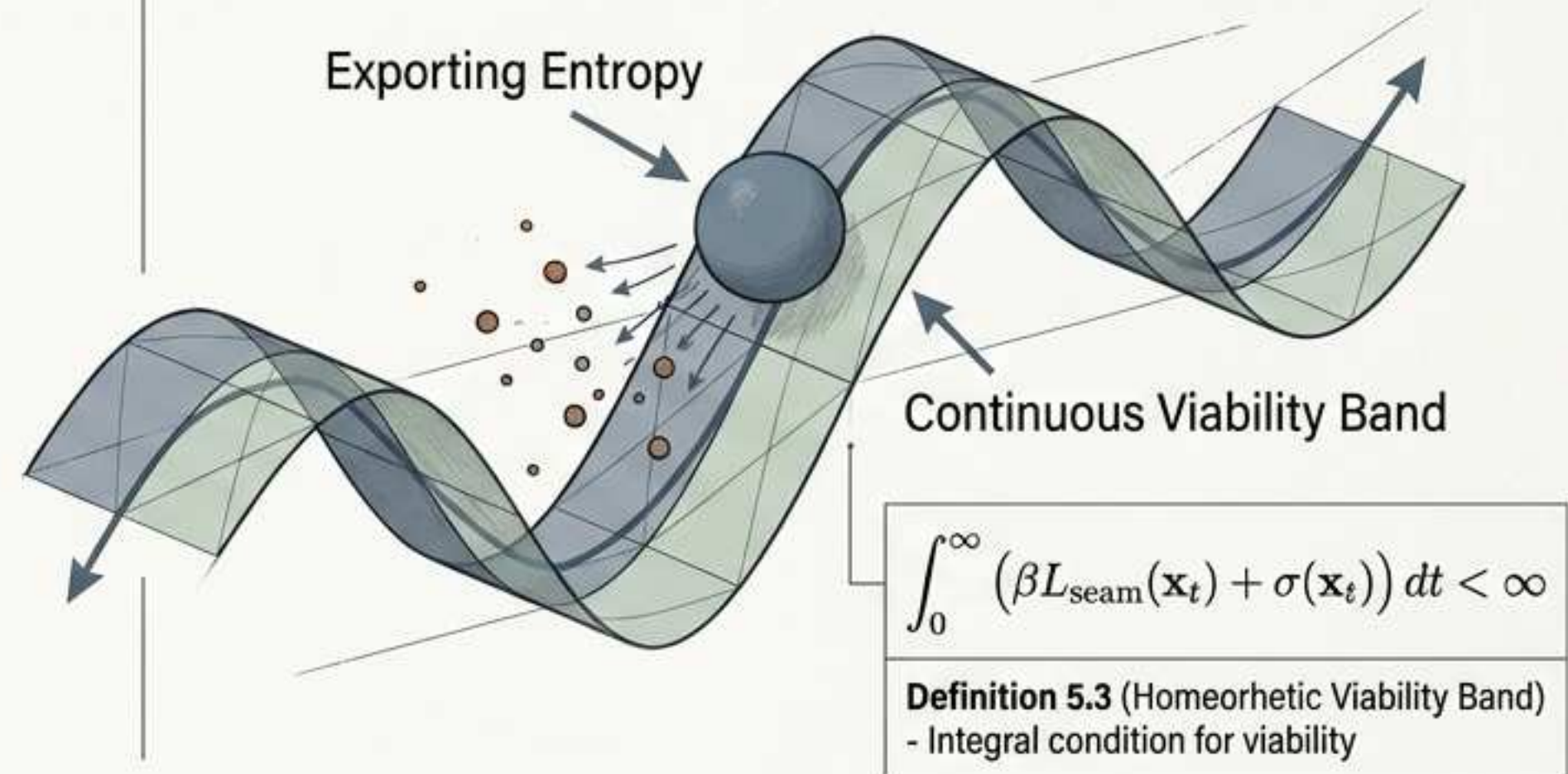


Viability demands stable dynamic trajectories, not static equilibrium.

Static Equilibrium



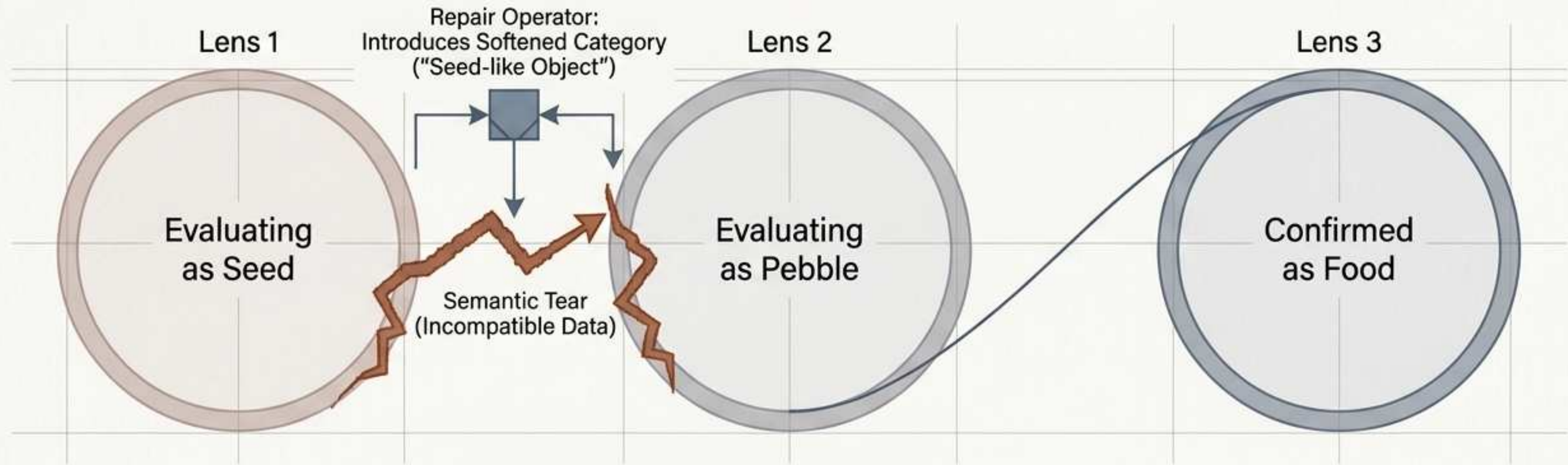
Homeorhetic Viability



Homeorhesis is the maintenance of a viable process, not the pursuit of a rest state.

The universe—and cognition—is *homeorhetically* stable in its entropy relaxation trajectory. To act intelligently is to sustain flows.

Local contradictions are stitched into global consistency without premature collapse.



A semantic tear occurs when local consistency fails on an overlap.

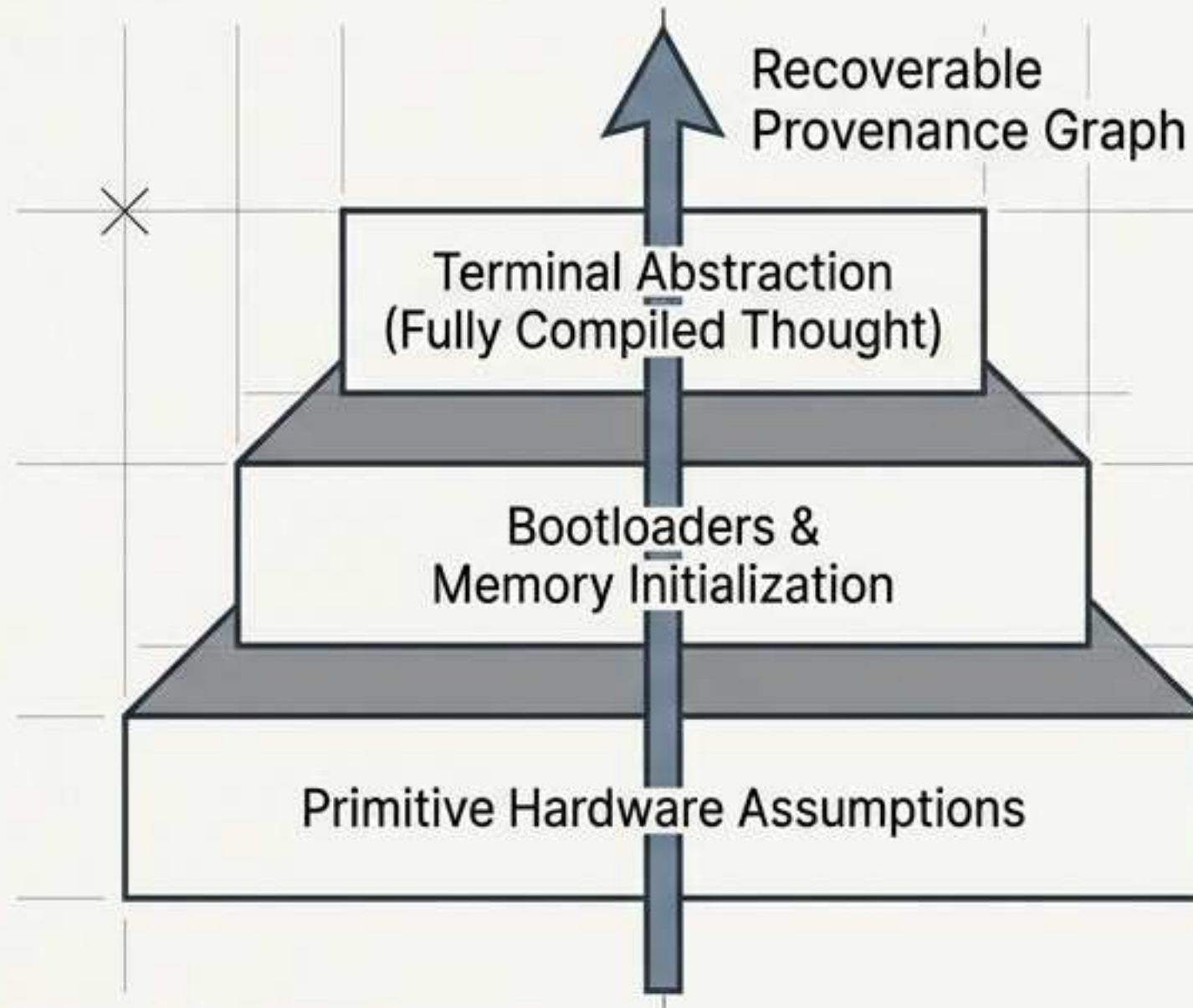
Repair introduces new objects or morphisms to trivialize the cocycle, deferring collapse until later windows arrive to confirm global context.

Definition 6.2 (Semantic Tear and Repair). A family of local sections $\{s_i \in S(U_i)\}$ is *glueable* if: (i) local consistency holds: $\rho_{U_i, U_j}(s_i) \cong \rho_{U_j, U_i}(s_j)$ for all overlapping pairs; and (ii) there exists a global extension $s \in S(\bigcup_i U_i)$ restricting to each s_i .

A *semantic tear* occurs when local consistency fails on some overlap $U_i \cap U_j$: the restriction of s_i and the restriction of s_j to their common domain are not isomorphic. A *repair* is the introduction of new objects or morphisms into $S(U_i)$ or $S(U_j)$ such that glueability is restored.



Meaning relies on the transparent preservation of the conceptual boot sequence.



An operating system does not appear instantaneously from nothing. Lower-level constraints must be stabilized before higher-level ones can be evaluated.

Erasing the ordered initialization history transforms an inspectable object into an opaque, dangerous proxy.

In an increasingly compressed world, intelligence is sustained by keeping the dependency graph explicitly recoverable.

Definition 2.8 (Boot Sequence). A boot sequence is the ordered set of constraint-stabilization events necessary to make a higher-level abstraction operationally meaningful. Lower-level constraints must be stabilized before higher-level ones can be evaluated. Erasing the boot sequence—presenting only the terminal abstraction without the ordered initialization history—transforms an inspectable object into an opaque, dangerous proxy.